

Similarity of spatial figures

One scale factor, three consequences — lengths by k , areas by k^2 , volumes by k^3 .

LESSON 13

1 × 40 MIN

GEOMETRY 11, §8

IGCSE E4.5

LESSON CLOCK

8'

14'

12'

8'

3'

What similarity means for solids	8 min
The three rules	14 min
Working backwards	12 min
Applications	8 min
Homework	3 min

LEARNING OBJECTIVES

- Decide whether two solids are similar.
- Apply the k , k^2 , k^3 rules to lengths, areas and volumes.
- Work backwards from an area or volume ratio to the scale factor.
- Solve problems on models, maps and scaled containers.

TEXTBOOK

National	pp. 71–76
Cambridge	Core & Extended, pp. 367–372
Workbook	Exercise 8.1

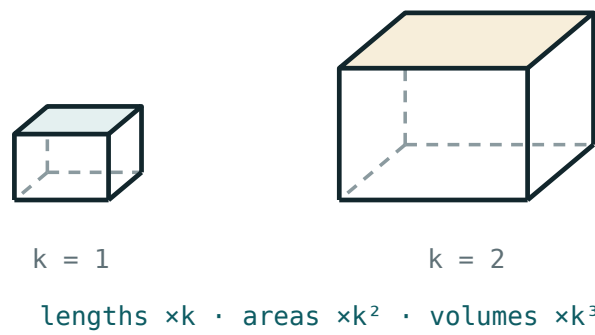
01

Explanation

Similar solids

DEFINITION

Two solids are **similar** when one is an enlargement of the other: all corresponding lengths are in the same ratio k , and all corresponding angles are equal.



Two cuboids in the ratio 1 : 2. Every edge doubles; the surface quadruples; the volume is eight times.

All cubes are similar. All spheres are similar. Two cuboids are similar only if their three edge ratios all agree — a $2 \times 3 \times 4$ and a $4 \times 6 \times 8$ box are similar; a $2 \times 3 \times 4$ and a $4 \times 6 \times 9$ are not.

The three rules

$$\frac{L_2}{L_1} = k \cdot \frac{S_2}{S_1} = k^2 \cdot \frac{V_2}{V_1} = k^3$$

K	LENGTHS	AREAS	VOLUMES
2	$\times 2$	$\times 4$	$\times 8$
3	$\times 3$	$\times 9$	$\times 27$
$\frac{1}{2}$	$\times \frac{1}{2}$	$\times \frac{1}{4}$	$\times \frac{1}{8}$
1.5	$\times 1.5$	$\times 2.25$	$\times 3.375$

THE COMMONEST ERROR IN THE WHOLE TOPIC

Doubling the dimensions does **not** double the volume — it multiplies it by eight. A model car at 1 : 20 needs $\frac{1}{8000}$ of the paint volume, not $\frac{1}{20}$.

Working backwards

Given a ratio of areas or volumes, recover k by taking the root:

$$k = \sqrt{\frac{S_2}{S_1}} \text{ or } k = \sqrt[3]{\frac{V_2}{V_1}}$$

Two similar cones have volumes 54 and 128 cm³. Then $k = \sqrt[3]{\frac{128}{54}} = \sqrt[3]{\frac{64}{27}} = \frac{4}{3}$,
so the heights are in the ratio 3 : 4 and the surface areas in the ratio 9 : 16.

PRACTICAL CONSEQUENCES

A model at 1 : 50 has $\frac{1}{2500}$ of the surface and $\frac{1}{125000}$ of the volume. This is why large animals have thick legs: mass grows as k^3 but bone cross-section only as k^2 .

02 Worked examples

EXAMPLE 1 Two similar cylinders have radii 3 cm and 5 cm. The smaller has volume 90 cm³. Find the larger.

$$① \quad k = \frac{5}{3}$$

$$② \quad k^3 = \frac{125}{27}$$

$$③ \quad V = 90 \times \frac{125}{27}$$

$$④ \quad = 416.7$$

ANSWER

$\approx 417 \text{ cm}^3$

EXAMPLE 2 Two similar solids have surface areas 48 and 108 cm^2 . Find the ratio of their volumes.

$$\textcircled{1} \quad k^2 = \frac{108}{48} = \frac{9}{4}$$

$$\textcircled{2} \quad k = \frac{3}{2}$$

$$\textcircled{3} \quad k^3 = \frac{27}{8}$$

ANSWER

$27 : 8$

EXAMPLE 3 A model aeroplane is built at $1 : 40$. The real wing area is 32 m^2 . Find the model's wing area in cm^2 .

$$\textcircled{1} \quad k = \frac{1}{40}, k^2 = \frac{1}{1600}$$

$$\textcircled{2} \quad 32 \div 1600 = 0.02 \text{ m}^2.$$

$$\textcircled{3} \quad 0.02 \times 10000 = 200 \text{ cm}^2.$$

ANSWER

200 cm^2

03 Interactive model

Build two cubes of side 2 cm and 4 cm and count how many small ones fill the large one.

Scaling up and down

Doubling every length multiplies the volume by:

Doubling every length multiplies the surface area by:

Volumes in the ratio 27 : 64 give lengths in:

Areas in the ratio 9 : 25 give volumes in:

3:5

9:25

27:125

81:625

A 1:100 model has what fraction of the volume?

$\frac{1}{100}$

$\frac{1}{10000}$

$\frac{1}{1000000}$

$\frac{1}{1000}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Similar solids	O'xshash jismlar	Подобные тела
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Ratio of areas	Yuzalar nisbati	Отношение площадей
Ratio of volumes	Hajmlar nisbati	Отношение объёмов
Corresponding lengths	Mos uzunliklar	Соответственные длины
Model (scale model)	Model	Модель
Enlargement	Gomotetiya	Гомотетия
Linear dimension	Chiziqli o'lcham	Линейный размер

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Similar solids have equal:

volumes

areas

angles

lengths

Areas scale by:

k

k^2

k^3

$2k$

Volumes scale by:

k

k^2

k^3

$3k$

All spheres are:

congruent

similar

neither

equal in volume

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $k = 2$; area factor

ANSWER 4

2. $k = 2$; volume factor

ANSWER 8

3. $k = 3$; volume factor

ANSWER 27

4. $k = \frac{1}{2}$; volume factor

ANSWER $\frac{1}{8}$

5. Volumes 8 : 27; length ratio

ANSWER 2 : 3

6. Areas 4 : 9; length ratio

ANSWER 2 : 3

7. Are all cubes similar?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. Radii 3 and 5, smaller volume 90; larger volume

ANSWER ≈ 417

2. Areas 48 and 108; volume ratio

ANSWER $27 : 8$

3. Volumes 54 and 128; length ratio

ANSWER $3 : 4$

4. $1 : 40$ model, real wing area 32 m^2 ; model area

ANSWER 200 cm^2

5. A cone is cut halfway up. Volume ratio of small cone to whole

ANSWER $1 : 8$

6. Same cut: ratio of small cone to frustum

ANSWER $1 : 7$

7. Two similar tins: heights 10 and 15, smaller holds 400 ml; larger

ANSWER 1350 ml

Hard · stretch and reason

7 PROBLEMS

1. A statue is 1:6 scale and needs 0.5 litres of paint. The real one needs?

ANSWER 18 litres

2. Two similar solids differ in volume by 117 cm^3 with $k = \frac{4}{3}$. Find both volumes

ANSWER 81 and 192

3. A cone of height 12 is cut at height 8. Volume ratio of the two parts

ANSWER 8 : 19

4. Why do large animals have proportionally thicker legs?

ANSWER Mass grows as k^3 , bone strength only as k^2

5. A 1:50 model has surface area 80 cm^2 . Find the real area in m^2

ANSWER 20 m^2

6. Two similar cylinders have total surface areas 150π and 600π . Find the volume ratio

ANSWER 1 : 8

7. Show that if two solids have equal volumes and are similar they are congruent

ANSWER $k^3 = 1 \Rightarrow k = 1$

07 Homework

Homework — 4 tasks

State k explicitly before using k^2 or k^3 .

- Two similar cones have heights 6 and 9 cm. The smaller has volume 80 cm^3 . Find the larger.
- Two similar solids have surface areas 75 and 300 cm^2 . Find the ratio of their volumes.
- A model bridge is 1 : 200. The real deck area is 4000 m^2 . Find the model's deck area in cm^2 .
- Explain, using k^2 and k^3 , why a doubled animal would break its own legs.